



Beyond Input Attribution: A Hands-On Tutorial to Concept-Based Explainable AI and Mechanistic Interpretability

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KDD 2025, Toronto - August 4th, 2025



mechanistic
interpretability

mechanistic interpretability

=

reverse engineering of neural networks
into human-understandable algorithms
and concepts

Feature #34M/31164353 Golden Gate Bridge feature example

The feature activates strongly on English descriptions and associated concepts

in the Presidio at the end (that's the huge park right next to the Golden Gate bridge), perfect. But not all people

repainted, roughly, every dozen years." "while across the country in san francisco, the golden gate bridge was

it is a suspension bridge and has similar coloring, it is often compared to the Golden Gate Bridge in San Francisco, US

They also activate in multiple other languages on the same concepts

ゴールデンゲートブリッジは、アメリカ西海岸のサンフランシスコ湾に接するゴールデンゲート湾

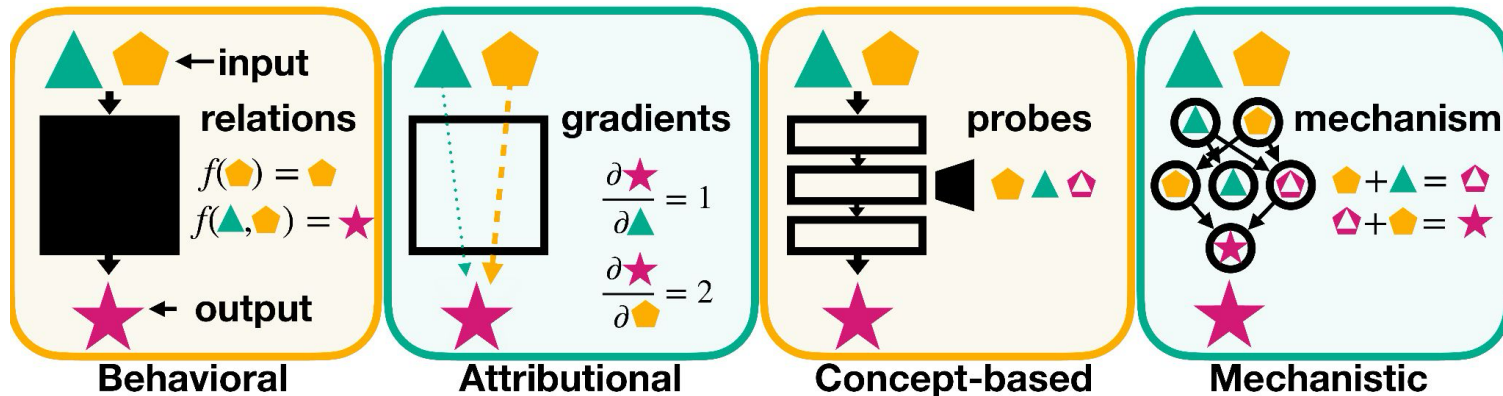
골든게이트교는 골문교는 미국 캘리포니아주 골든게이트역에 위치한 교이다. 골든게이트교는 캘리포니아주 샌프란시스코

мост Золотые Ворота — висячий мост через пролив Золотые Ворота. Он соединяет город Сан-Фран

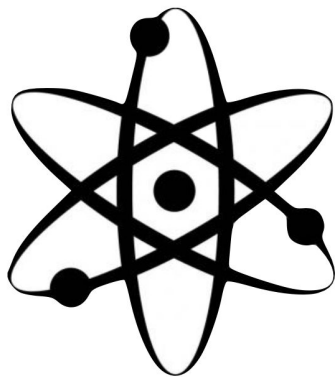
And on relevant images as well



<https://transformer-circuits.pub/2024/scaling-monosemanticity/>



features (or latents) indicate
the **minimal** units of representation that
cannot be reduced (disentangled) into simpler factors



monosemantic vs polysemantic features & the role of superposition

monosemantic features

- Each neuron/dimension responds to a **single clear concept**
- Example: one neuron fires only on "car" concept

🟢🔍 easy to interpret

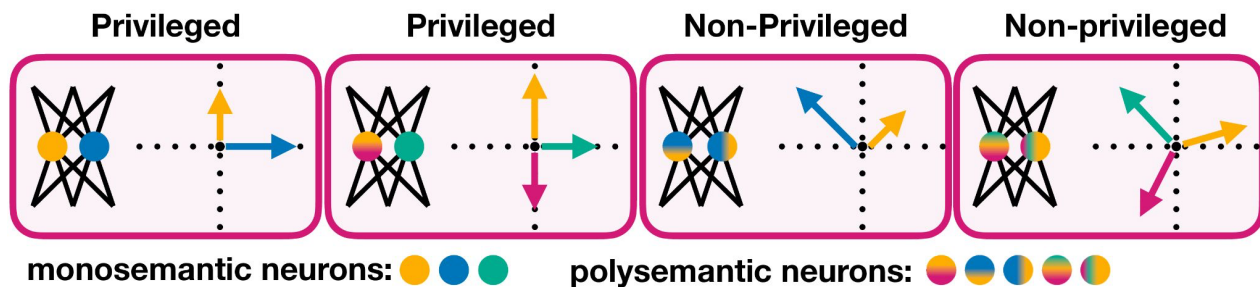
polysemantic features

- A neuron responds to **multiple unrelated concepts** (e.g. cats and cars)

🔴🧩 hard to interpret

superposition

- Models often represent **more independent features than they have neurons**
- Achieved by **packing multiple concepts into the same neuron** via linear combinations (quasi-orthogonal representations)
- Leads to **polysemanticity**



core methods

Observation



**Structured
Probes**

Logit Lens

**Sparse
Autoencoder**

concept-based,
supervised

unsupervised

Intervention

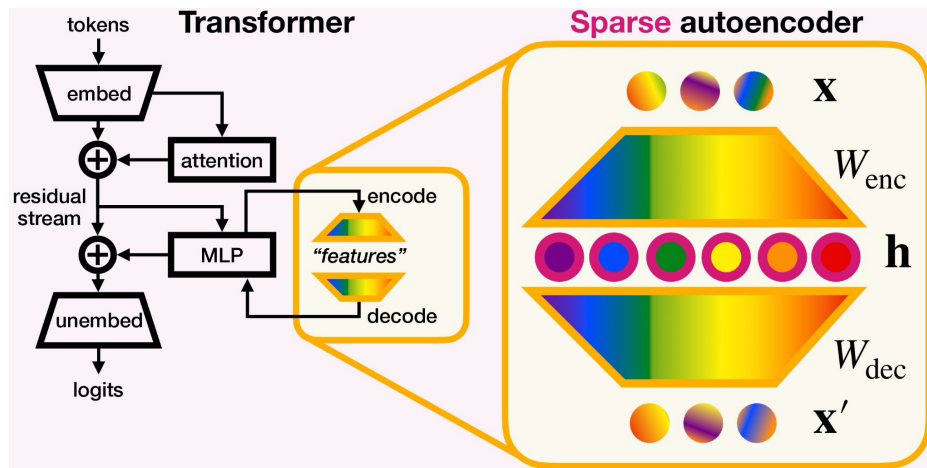


**Activation
Patching**

**Attribution
Patching**

**Causal
Scrubbing**

sparse autoencoders



Hands on!